Introduction to

Competitive Programming

Arman Ferdous

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What is Competitive Programming?

- Contestants compete to solve some computable problems faster than others
- What is a non-computable problem?
- Why solve problems that are already solved?
- How your solutions are judged?
- Often the best solution is not the concern, rather one that fits the constraints.
- You won’t be building any complex software applications.

An imaginary story between Alice & Bob (1)

- Alice and Bob will hangout tomorrow at the UIU CSE festival.
- Alice needs to get ready before leaving.
- She needs to
  - Charge her phone
  - Download the tickets
  - Finish an assignment
  - Email few faculties
  - Do her morning activities

(2)

- Alice needs to check how to reach her destination fast.
- So she checks Google Maps.

(3)

- Bob on the other hand had trouble last night.
- He had several assignments and a project he was working on.
- He had to figure out how he can spend his time on each task so that the most amount of work is done.

(4)

- Bob wanted to bring some snacks before arriving.
- But his backpack is already quite heavy from his laptop and equipments.
- So he needs to keep the snacks within a weight limit.

(5)

- Alice & Bob have reached the festival!!!
- They want to visit all the stalls and attractions.
- So they plan how to do it with the minimal effort.

(6)

- They reach a stall where you can play a game
- It is a spinning wheel which has different awards.
- Bob started counting how many sectors of the wheel contained a good outcome for them. He found that 8 out of 50 sectors had good rewards.

(7)

- Due to the huge crowd, the internet got slow and unusable :(
- Alice and Bob had received walkie talkie when they arrived.
- When trying to use it they find out that it can only make beep sounds and nothing else :(
- So they have to devise a plan to communicate with each other. They only need to support several keywords.

(8)

- The organizers need to connect the stalls via fiber cables to give them internet access.
- As long as every stall is connected to the network it will work.
- So they try to minimize the amount of fiber cables to join all stalls together.

(9)

- The festival is almost over. Alice and Bob are helping in cleanup duties.
- They were given some packages that they need to put inside large containers.
- The containers will be taken away via trucks afterwards.
- They need to use the minimum number of containers to pack all their packages inside.

So how many classical problems did you find?

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(1) Topological Sort
(2) Shortest Path
(3) Task Scheduling
(4) 0-1 Knapsack
(5) Travelling Salesman
(6) Games with probability & expected values
(7) Lossless Data Compression
(8) Minimum Spanning Tree
(9) Bin Packing

Algorithms/DS are connected to our daily lives

- So it makes sense to study them
- One of the best ways to keep your algorithmic senses sharp is through CP.
- It can also make you a better coder.

The Big Competitions

- IOI: For school & college students.
- ICPC: For university students.





The ICPC Environment

- Teams of 3 with 1 PC
- 10 - 15 problems, 5 hours
- No internet access
- You get a 25 page reference doc
- Each solve gives a balloon



How to reach the ICPC World Finals?

1. ICPC Dhaka Regional Preliminary Contest
2. ICPC Dhaka Regional Contest
3. ICPC Asia West Continent Championship
4. ICPC World Finals!



What about other competitions?

- Regular Inter University Programming Contests are held now!
- Yesterday, we had an IUPC at SUST where 1 team from UIU went
- Slots…

We need a strong team with solid pipeline

- Once a team starts getting up the ranks we will be gaining more slots.
- Some examples: IIUC, Green University.

How do we measure our abilities?

- Online contests!
- Less informal than onsite ones.
- Shorter format.
- Rating Systems.
- Cheaters







Example

- https://codeforces.com/topic/137804/en14

Our Goals

- Reach a point where we can start to believe that it is possible to do X
- Actually obtain X
- Repeat

Let’s solve some problems

Watermelon

- Two friends want to divide a watermelon that weights w ($1 \leq w \leq 100$) kilograms.
- They want to divide the watermelon in such a way that each of the two parts weighs an even number of kilos, but it is not necessary for the parts to be equal.
- Given w determine whether such division is possible or not.
- Print ”YES” or ”NO”.

Examples

input
8
output
YES

Verdicts

- Accepted: You solved the problem.
- Wrong Answer: Your code produced incorrect output.
- Time Limit Exceeded: Your code did not finish within the time limit.
- Runtime Error: Your code crashed while running.

There are more, but this much is enough for now.

Measuring how long your code will take

- A computer used for judging can perform around 10^8 primitive operations in one second.
- So we calculate how many operations our code will require.
- Time Complexity!

Example 1

```
// Take n, m input from the user.
sum = 0;
for (int i = 0; i < n; ++i)
    for (int j = 0; j < m; ++j)
        for (int k = 0; k < 5; ++k)
            sum += i * j * k;
```

Example 2

```
# A python example
n = int(input())     # length of array
arr = [int (x) for x in input().split()]

for i in range(n):
    print(sum(a) - a[i])
```

Example 3

```
# Take a positive integer x from the user.
while (x) {
    if (x % 2 == 0) x /= 2;
    else x--;
}
```

What type of skills to train

There are 3 types of skills one has to train

a) Observational skills
b) Theoretical skills
c) Coding skills

Observational skills are the hardest to train and take the longest to improve.

Water Bottle Puzzle

There is a 3L bottle, and a 5L bottle. You have an infinite supply of water. Can you measure 4L of water? The bottles have no marks on them, so your only options are to fill them up entirely/empty them out/switch from one bottle to another.

Prisoner’s Hat Puzzle

Alice and Bob are in prison and the warden plays a game with them. The warden puts on a hat (White/Black) on each of their heads while they were not looking.

Now Alice and Bob are in front of each other and can see the other person’s hat, but can’t see their own.

They must guess together what their own hat color is. If either one of them is correct the warden will let them go. Find a strategy for Alice & Bob.

Takeaway

- Although they seem difficult, if you start attempting to solve the ideas will flow.
- If you don’t try at all, the options will always seem overwhelming. But if you calmly take it one step at a time, you can find the solutions without even realizing.
- These problems are ”Linear” in a sense - ”Following the most obvious route leads you to the solution”.

Ants on a Rod

- N ants are on a 1m length rod. Each ant has a position $0 \leq x_i \leq 1$ which is a real value, and a direction it is facing (either L or R).
- All ants start moving at speed 1m/s, at time = 0.
- When two ants collide, they change direction and start moving in the opposite direction instantly. When an ant reaches one of the ends of the rod, it falls off.
- Find the time it takes for all ants to fall off.

Pattern Hunting (LightOJ 1136)

There is a sequence 1, 12, 123, ..., 123456789, 12345678910, ...

Given A and B, find the number of integers divisible by 3 from the Ath term to the Bth term.

T (≤ 10000) test cases.

1 ≤ A ≤ B < 2^31.

Codechef PDEL

A string is palindrome if it reads the same from backwards. You are given a string S. In a single move you can delete any non-palindromic substring from it (can't delete the whole string).

Find the minimum number of moves to turn S into a palindrome, or report it is impossible to do so.

Example: racecar → 0

Example: modem → 1

Example: my → impossible.

Limits: N ≤ 2 × 105.

What should you start doing immediately?

- Do online contests. Weekly 1-2.
- Try to solve the problems even after the contest ends.
- Read some books for algorithm/ds knowledge.

Free Resources

- USACO Guide: Structured step by step things to learn
- CP Algorithms: Website which explains pretty much every algo/ds for contests.
- CP Handbook: Free book for CP, available online
- Codeforces blog posts.

Other Books

- Programming Contest: Data Structure & Algorithms - Mahbubul Hasan Shanto.
- Competitive Programming Book 4 - Steven Halim

QnA

My email: arman.ferdous@g.bracu.ac.bd

Thank you!